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Climate change impacts on European crop yields: Do we need to consider nitrogen limitation?

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ABSTRACT

Global climate impact studies with crop models suggest that including nitrogen and water limitation causes greater negative climate change impacts on actual yields compared to water-limitation only. We simulated water limited and nitrogen–water limited yields across the EU-27 to 2050 for six key crops with the SIMPLACE<LINTUL5, DRUNIR, HEAT> model to assess how important consideration of nitrogen limitation is in climate impact studies for European cropping systems. We further investigated how crop nitrogen use may change under future climate change scenarios. Our results suggest that inclusion of nitrogen limitation hardly changed crop yield response to climate for the spring-sown crops considered (grain maize, potato, and sugar beet). However, for winter-sown crops (winter barley, winter rapeseed and winter wheat), simulated impacts to 2050 were more negative when nitrogen limitation was considered, especially with high levels of water stress. Future nitrogen use rates are likely to decrease due to climate change for spring-sown crops, largely in parallel with their yields. These results imply that climate change impact studies for winter-sown crops should consider N-fertilization. Specification of future N fertilization rates is a methodological challenge that is likely to need integrated assessment models to address.

1. Introduction

Growing world population and consumption, together with recent record high food prices, have led many to speculate on how food security will evolve in the future (Godfray et al., 2010). A complex mix of factors underlies food price variability, such as fluctuations in the price of oil, bio-energy policies, speculation in food markets, relative strength of currencies, agricultural and trade policies, growing global affluence and meat consumption, and climatic extremes (Westhoff, 2010). Climate change, declining expenditures for agricultural research, conflicts and health epidemics constitute additional challenges to future food security (Rosegrant and Cline, 2003). Nonetheless, crop productivity remains a key factor influencing food prices and food security. Hertel et al. (2010) found substantial differences in simulated crop commodity prices and poverty outcomes to 2030 due to climate change impacts on crop yields, depending on which crop-growth model result was used. They concluded that improved understanding of how climate change impacts crop yields needs to be a research priority.

Actual and future crop yields are determined by a combination of factors, among others those that determine potential yields (temperature, radiation, CO_2), limit these yields (water and nutrients) and further reduce it due to biotic (weeds, disease, pests) and abiotic (e.g. ozone, salinity) stresses (Neumann et al., 2010; van Ittersum et al., 2013). Process-based crop models are used to study how crop yields react to changes in climate, though they differ widely in their representation of various soil and crop processes (White et al., 2011). In general, their predictive power declines the more realized actual yields deviate from potential yields (van Ittersum and Rabbinge, 1997). While many crop models can consider basic management (e.g. irrigation and nitrogen (N) fertilization), large-scale studies typically neglect N stress and irrigation, assuming either all rainfed production or only potential yields, due to a lack of reliable data on crop specific N and irrigation application rates and timing (Reidsma et al., 2010). While the uncertainty with regard to actual management makes a case for focusing on simulating potential or water-limited yields, in a global analysis of crop yields to 2050, Rosenzweig et al. (2014)

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found that models that considered N-limitation simulate more negative climate change impacts compared to the models that do not. However, no analysis was offered to explain the mechanism behind this response, and no models were replicated both with and without nitrogen stress, limiting the comparability of the results for the two production situations.

Generally, approximating actual yields realized under future climates faces a twofold challenge: (1) projecting factors not considered by crop models and not addressed in this study—such as the use of improved cultivars, better weed, disease and pest control and (2) projecting those that are explicitly taken into account such as N application. However, for the latter, particular challenges refer to the understanding and consideration in a modelling approach of e.g. N application rates and how these will evolve over time and interact with climate change effects on yields. Analysis of past yield trends can be used to generate scenarios of how actual yields may improve based on the combined effect of a higher yield potential, such as by using new cultivars and by closing the yield gap with improved technology and crop management (Ewert et al., 2005). Such analysis can account for factors typically not considered in crop models. However, such purely data driven empirical methods are criticized for their lack of structural analysis (van Ittersum et al., 2013) and provide little explanation about why projected changes may occur.

One approach to specify future crop management such as N fertilization is to assume that current yield gaps remain constant in relative terms, and adjust N application from baseline levels by the factor that potential or water limited yields change in response to both climate and technology factors (Wolf et al., in press). In reality, N application rates depend on their marginal returns, and large scale estimation of the marginal returns would require iterations between large scale (preferably global) economic models of the agricultural sector, farming systems models and crop models, involving different steps of aggregating and disaggregating results between scales and models (Britz et al., 2012). Such an approach is currently not feasible for large scale studies. Furthermore, at national and global scales, clear trends in N use and nitrogen use efficiency can be discerned that vary in time and with the region, making specification of future N use more challenging (Lassaletta et al., 2014).

Within the context of limited data on current N use together with the need to make a number of assumptions to specify future nitrogen fertilization, the objective of this study is to assess how important consideration of N limitation is in climate impact studies for crops at the European scale representing different combinations of growing conditions and management intensities. A second objective is to estimate how crop N uptake may change under future climate change scenarios. The assessment was made with the SIMPLACE modeling framework (Gaiser et al., 2013), including, amongst others, the LINTUL5 crop growth model (Wolf, 2012). The LINTUL models have been used to assess water use under different irrigation regimes (Farre et al., 2000), crop nitrogen limitation resulting from different nitrogen fertilization rates in rice (Shibu et al., 2010), as well as in climate change impact assessment studies (e.g. van Ittersum et al., 2003; Angulo et al., 2013). The LINTUL models have been used with SIMPLACE in various studies at field (Gaiser et al., 2013), country and continental scales (Zhao et al., 2015a).

2. Materials and methods

2.1. Simulation setup

SIMPLACE<LINTUL-5, DRUNIR, HEAT> (Gaiser*et al.*, 2013; Zhao et al., 2015a Zhao et al., 2015a) was used to simulate daily growth and development of six crops (grain maize, potato, sugar beet, winter barley, winter rape seed and winter wheat) for the EU-27. Climate and soil inputs were provided at the level of simulation unit (intersection of agri-environmental zones (Metzger et al., 2005) and NUTS2 administrative zones) (referred to as agro-climatic zones in Angulo et al., 2013) across Europe (Fig. 1). Simulations for each crop were conducted twice: (1) water limited (WL) simulations in which water but not nitrogen limits crop growth; and (2) nitrogen–water-limited (NWL) simulations in which both water and nitrogen limit crop growth. In both cases, simulations were conducted for a historical period between 1982 and 2006 and a scenario period of 2040–2064, nominally referred to as baseline and 2050, respectively.

WL and NWL simulated yields considered the current irrigation fraction in each simulation unit. For each crop and each simulation unit, the respective WL and NWL yields were estimated by taking the weighted average of (1) yields simulated with no-water stress (representing full irrigation) and (2) yields simulated under rainfed conditions with water stress (Zhao et al., in press). The weighting was performed on the basis of irrigated vs total cropped area of each crop as reported in the global irrigation map (Siebert et al., 2005). The irrigated and total cropped areas are originally from the Eurostat database (2014) at the level of NUTS3, but were aggregated to the NUTS2 level by taking an area weighted averages. All runs for both WL and NWL simulations were performed consecutively for each year with annual re-initialization, as no spatial information on crop rotations was available.

2.2. Model description

SIMPLACE<LINTUL-5, DRUNIR, Heat> used in this study combines the LINTUL-5 crop growth model (Wolf, 2012), the water balance model, DRUNIR from the LINTUL-2 model (Spitters and Schapendonk, 1990) and a module to simulate the effects of high temperature on grain yield (Eyshi Rezaei et al., 2013) in the SIMPLACE modeling framework (Gaiser et al., 2013). The code for SIMPLACE and the model components used in this study can be assessed at: http://www.simplace.net/Joomla/index.php/documentation/simplace-framework-for-software-developers

In LINTUL5, daily temperature sums and crop-specific requirements are used to simulate the crop development stage (DVS) and progression from emergence (DVS = 0.0) to anthesis (DVS = 1.0) and maturity (DVS = 2.0). Additionally for the winter sown crops, preanthesis development is sensitive to a vernalization requirement and photoperiod. Leaf area index (LAI) expansion is exponential until DVS = 0.2 and LAI < 0.75 are reached, limited only by temperature. Later LAI growth depends on a number of factors including: daily increase in leaf biomass, specific leaf area, leaf death rates and nitrogen and water stresses. Beer's Law is used to calculate the quantity of intercepted photosynthetically active radiation as a function of LAI. The model uses the concept of radiaton use efficiency, which can vary throughout the crop's growth cycle and with daily average temperature and atmospheric CO_2 concentration, to determine daily biomass increment. Total biomass growth is limited by the most limiting of either water or N-stress. The effects of atmospheric CO_2 concentration on radiation use efficiency and transpiration rates are detailed in Zhao et al. (in press). Biomass is partitioned to roots, stems, leaves and storage organs (yield) as a function of DVS, water and N-stress. A one-layer tipping bucket approach is used to model the soil water balance and nutrient availability. The water stress factor is defined as the ratio of actual transpiration to potential transpiration with the Penman equation (1948) used to estimate potential crop evapotranspiration.

In LINTUL5, N uptake from the soil equals the smallest of either total crop N demand or total mineral N available from soil on that day. Subsequently, N is partitioned to leaves, stems and roots (stor-

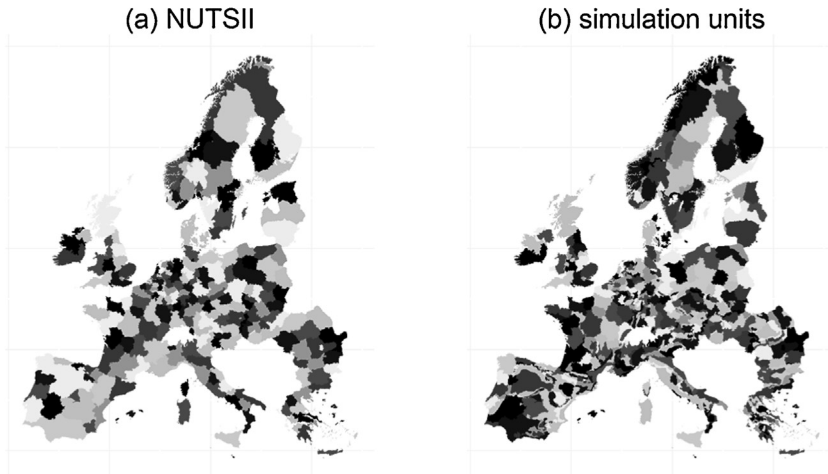

Fig. 1. Spatial resolution of (a) NUTSII (nomenclature of territorial units for statistics) regions (n = 240), that were the basis of historical yield and fertilizer information abstracted from the CAPRI database, and (b) the model simulation units (n = 534), defined by the spatial intersection of NUTSII zones and environmental zones as defined in Metzger et al. (2005).

age organs derive their N from translocation from the other crop organs). The amount of soil mineral N is determined by the effective fertilizer N application, the initial mineral N content and the soil N mineralization rate. N-uptake is reduced by water stress and ceases after a crop specific DVS. Daily crop N demand for all crop parts is determined by considering their respective maximum N amounts (product of maximum concentration and current dry matter content) which depends on crop DVS and their actual N content. The amount of N available for translocation to the storage organs from leaves, stems and roots depends on their actual N content and their required residual N amounts (product of residual concentration and mass). Maximum and residual N concentrations can vary with crop DVS. The actual translocation of N to the storage organs depends on crop DVS, is moderated by a time constant and is limited by both the demand and supply. N stress due to shortage of N is quantified using the Nitrogen Nutrition Index (NNI) calculated as the difference between actual and residual crop nitrogen contents divided by the difference between optimal and residual crop nitrogen contents. NNI affects leaf area expansion, biomass growth, leaf death rate and partitioning of biomass to various organs (Wolf, 2012)

The LINTUL models have been used to assess water use under different irrigation regimes (Farre et al., 2000), crop nitrogen limitation resulting from different nitrogen fertilization rates in rice (Shibu et al., 2010), as well as in climate change impact assessment studies (e.g. van Ittersum et al., 2003; Angulo et al., 2013). The LINTUL models have been used with SIMPLACE in various studies at field (Gaiser et al., 2013), country (Eyshi Rezaei et al., 2015) and continental scales (Zhao et al., in press).

Phenology (temperature sums from emergence to anthesis, TSUM1, and from anthesis to maturity, TSUM2) was calibrated for each simulation unit and crop using observations of sowing, anthesis and harvest dates. When phenology observations were not available, sowing dates were estimated from a similar crop in the same environmental zone and the LINTUL5 default ratio of TSUM1 to TSUM2 was used when anthesis observations were missing. As data is not available at a European scale to specify cultivars (Therond et al., 2011), all other parameters related to crop growth were the LINTUL5 default values as reported in Boon-Prins et al. (1993). Nitrogen concentration of various plant tissues were kept constant at the LINTUL5 default values with increasing CO_2 concentrations. It is assumed that these CO_2 effects on N concentration is negligible compared to other uncertainties (e.g. differences between crop varieties, differences in fertiliser N management).

2.3. Climate, soil, and phenology data

Baseline climate data (Van der Goot and Orlandi, 2003) included daily minimum and maximum temperature, solar radiation, precipitation, and wind speed and was obtained from the SEAMLESS database (van Ittersum et al., 2008). Future climate data were generated by adding climate deltas from monthly GCM model outputs to the baseline climate data. Three climate scenarios from the Special Report on Emissions Scenarios (SRES) (Nakicenovic and Swart, 2000) were considered: the B1 with GCM MIROC3.2, and B2 and A1B with a 15-member GCM ensemble as downloaded from the Data Distribution Centre of the IPCC (2010). The B1 scenario, or baseline scenario, is characterized as a continuation of current trends towards increasing globalization, whereas the B2 scenario is characterized by low economic growth and the development of regions. In contrast, the A1B scenario has stronger economic growth than the others and a continuation of globalization trends with a balance between fossil and alternative fuels scenarios (Parry, 2007). Atmospheric CO_2 concentrations of 532, 487 and 478 ppm were used for the A1B, B1 and B2 scenarios, respectively. While the IPCC Fifth Annual Assessment Report (Stocker et al., 2013) uses representative concentration pathways (RCPs) and no longer the SRES scenarios of the Third and Forth Assessment Reports, this study maintained the use of the older SRES scenarios. It was decided to use the SRES scenarios as this study was part of a larger European-wide integrated assessment performed for the DG Agriculture in Brussels involving a number of models and modelling groups which was started before the RCPs were developed (Ewert et al., 2011). Wolf et al. (in press) describe the relationship of the scenarios used in this study to the RCPs. Absolute monthly deltas for minimum and maximum temperature and relative deltas for precipitation and radiation were calculated directly from the GCM outputs by comparing future and baseline results. Near surface windspeed from the model outputs was calculated by taking the square root of the sum of squared zonal and meridional components before deltas were calculated. Daily deltas were interpolated from the monthly values using cubic splines (Ewert et al., 2011). The changes in daily maximum temperature, radiation and precipitation relative to baseline conditions are shown in Fig. 2.

Soil–water holding characteristics needed for use in SIMPLACE<LINTUL5, DRUNIR, HEAT> are water content at saturation, field capacity, permenant wilting point, air day and a threshold for damage due to lack of oxygen, and the maximum soil rooting depth. These values were taken from the SEAMLESS database (van

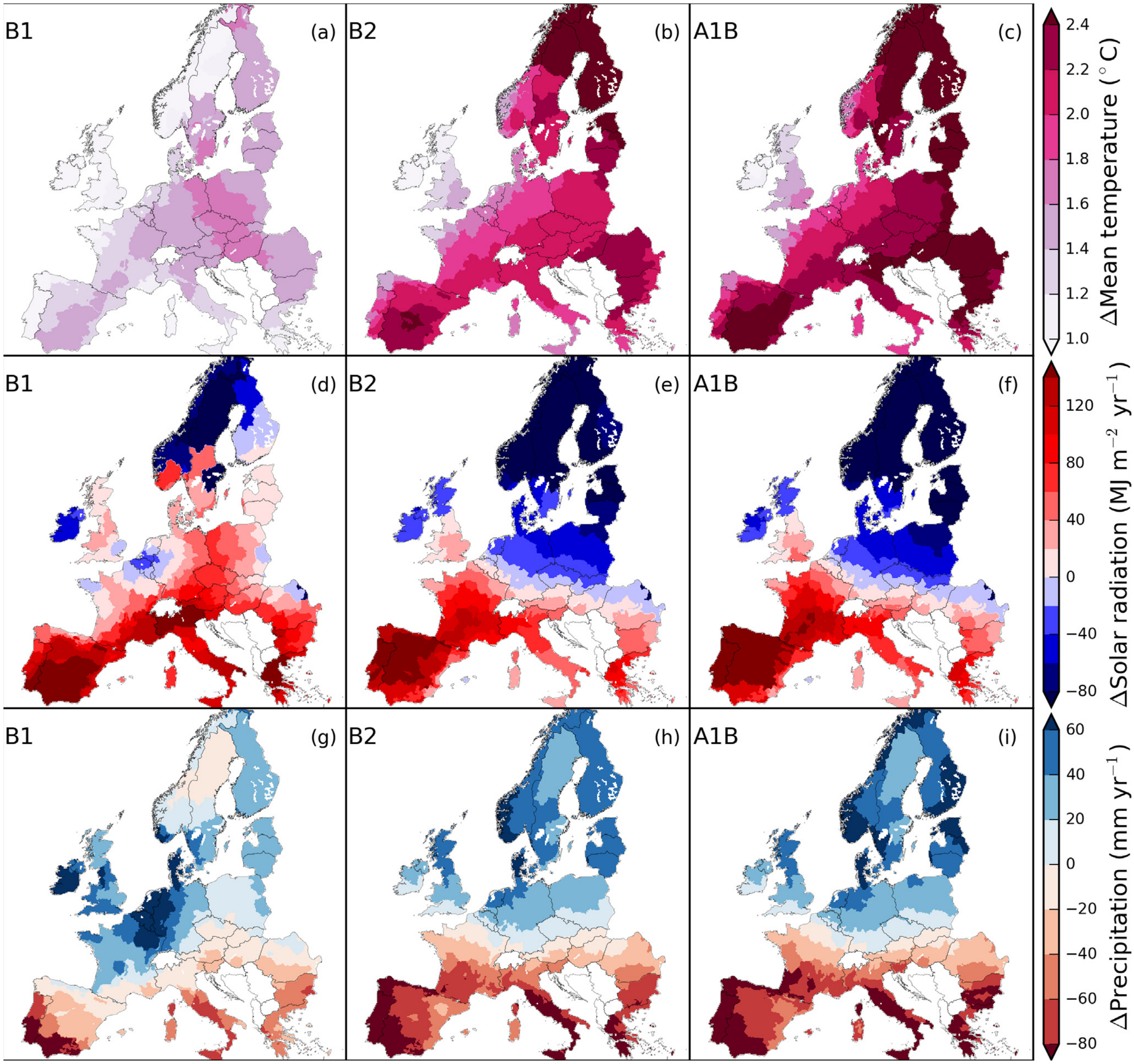

Fig. 2. Changes in mean daily temperature (a–c), and annual global solar radiation (d–f) and precipitation (g–i) sums for the three SRES climate change scenarios, B1 (a, d, g), B2 (b, e, h) and A1B (c, f, i).

Ittersum et al., 2008; Janssen et al., 2009), in which the majority soil type was selected for each agri-environmental zones (Ewert et al., 2011). Soil types are differentiated on the basis of the organic carbon content in topsoil (Hazeu et al., 2010). Observations of sowing, anthesis and harvest (assumed equal to maturity) dates were used from the JRC/MARS database (http://mars.jrc.ec.europa.eu/mars). The observations were aggregated to the agri-environmental zones. Historical data on grain yield were available from the Eurostat database (EUROSTAT, 2014) for the period from 1982–2006, as processed for in the CAPRI database (Britz et al., 2007). Simulations were conducted in each simulation unit in which observed historical yields and nitrogen fertilizer rates were available (Fig. 1).

2.4. Nitrogen inputs

In the baseline period (1982–2006) simulations, total nitrogen available to the crop in SIMPLACE<LINTUL5, DRUNIR, HEAT> (Fig. S1) was the sum of (1) initial soil mineral N, and (2) inputs determined for each crop from historical data on annual nitrogen use in the CAPRI database (Britz et al., 2007; Leip et al., 2011) from various sources (mineral fertilizer, manure and crop residues) for each NUTS2 region (http://epp.eurostat.ec.europa.eu/portal/page/portal/nuts_nomenclature/introduction).

The initial mineral N content was assumed to equal to 40 kg ha^{-1} for each simulation unit. Mineral nitrogen in cropland soils is highly variable in space and time. Therefore, in each country recommendation systems are established to advice farmers regarding expected mineral N content in the soil before planting. According to estimates for Germany (NRW, 2014; Wendland and Kavka, 2014a,b) and Spain (IDEA, 2007), 40 kg ha^{-1} is a reasonable average over different soil types and pre-crops for spring and winter crops. As the mineral N content varies from plot to plot and year to year, it is not feasible to make spatial and temporal differentiation of soil mineral N content in a European wide study over a period of several years.

To determine the total annual N inputs from manure, information on the total herd size (summed for cows, pigs and chickens) at the NUTS0 (national) level and the daily manure production was used to estimate the relative contributions of different manure types and the total manure applications per hectare. The mineral N content in manure was used as a proxy for plant available nitrogen (Chadwick et al., 2000). N losses due to ammonification were assumed to be equal to the fraction of the organic-N in the manure which comes available during the year. In CAPRI, NH_3 losses were reported as an aggregate for all crops on arable land, and the assumption was made that the relative losses from manure were constant across crops. Plant available N in crop residues was determined using an average annual N-mineralization fraction of 0.20 for crop residues. This mineralization rate is for biomass residues with nitrogen concentrations of 20 g kg^{-1} biomass. This value approximately corresponds to high quality cereal stover, which account for upwards of 60% of crops in common rotations. As SIMPLACE<LINTUL5, DRUNIR, HEAT> does not consider leaching, the constant mineralization rate is thought to be a satisfactory approximation. Mineral fertilizer-N was taken directly from the CAPRI database. To calculate future nitrogen fertilization rates the arbitrary assumption was made that the management intensity would not change, i.e. the yield gap due to nitrogen would stay the same. The future nitrogen (N_{2050}) was determined by changing fertilization levels in the baseline (N_{2004}) by the same factor that WL yields changed (Δ_{WL}) due to climate change and CO_2 such that:

$$N_{2050} = N_{2004}(1 + \Delta_{WL}/100).$$

Therefore, if Δ_{WL} was negative, less nitrogen would be applied in the future whereas if Δ_{WL} was positive more nitrogen would be applied in the future. In all cases, all N-fertilizer was applied at sowing.

2.5. Climate change impacts

The relative change in yields over time, Δ_{WL} and Δ_{NWL} for water-limited and nitrogen–water-limited yields, respectively were calculated as:

$$\Delta_{WL} = \frac{Y_{WL,2050} - Y_{WL,2004}}{Y_{WL,2004}} \times 100\% \tag{1}$$

and

$$\Delta_{NWL} = \frac{Y_{NWL,2050} - Y_{NWL,2004}}{Y_{NWL,2004}} \times 100\% \tag{2}$$

where $Y_{WL,2050}$ is the mean water limited simulated yield over the scenario period (2041–2064) and $Y_{WL,2004}$ is the water limited yield estimated for 2004 from the regression through simulated yields over the baseline period (1984–2004). This was not always equal to the mean yield, as there was sometimes a trend in the nitrogen inputs that translated into a trend in simulated yields in the baseline period. In the future scenario period, there was no trend in the nitrogen data or yields. The difference between the relative yield changes for WL and NWL simulations, $\Delta\Delta$ was determined as:

$$\Delta\Delta = \Delta_{WL} - \Delta_{NWL} \tag{3}$$

As NU_{WL} is simulated with no nitrogen limitation, it is numerically equal to the N demand of the crop. NU_{NWL} is simulated considering nitrogen available to the crop. The relative change in NU, $\Delta NU_{,WL}$ and $\Delta NU_{,NWL}$ for water-limited and nitrogen–water-limited simulations, respectively, were calculated as:

$$\Delta_{NU,WL} = \frac{NU_{WL,2050} - NU_{WL,2004}}{NU_{WL,2004}} \times 100\% \tag{4}$$

and

$$\Delta_{NU,NWL} = \frac{NU_{NWL,2050} - NU_{NWL,2004}}{NU_{NWL,2004}} \times 100\% \tag{5}$$

Table 1
European aggregate relative yield changes between 2004 and 2050 for three climate scenarios estimated using WL and NWL simulated yields.

Crop	Relative yield changes (%)					
	A1B scenario		B1 scenario		B2 scenario	
	Δ_{WL}	Δ_{NWL}	Δ_{WL}	Δ_{NWL}	Δ_{WL}	Δ_{NWL}
Grain maize	−4	−4	1	2	−5	−5
Potato	−5	−6	1	−1	−8	−9
Sugar beet	12	11	11	11	4	3
Winter barley	19	6	14	3	11	1
Winter rapeseed	19	15	14	10	11	8
Winter wheat	14	0	13	4	7	−4

Table 2
Observed and simulated European aggregate baseline yields in 2004 estimated from WL and NWL simulations with a NUTS2 specific yield correction factor included.

Crop	2004 baseline yields (t ha^{-1})		
	Obs	WL Sims	NWL Sims
Grain maize	6.3	9.4	6.7
Potato	5.1	9.0	6.4
Sugar beet	10.9	14.0	12.7
Winter barley	3.5	4.7	3.0
Winter rapeseed	2.2	4.1	3.0
Winter wheat	4.0	6.0	4.2

3. Results

3.1. Climate change and elevated CO_2 concentration impacts European crop yields

The combined impact of climate change and elevated CO_2 concentrations on the six crops for three emission scenarios averaged across Europe were expressed as relative yield changes between 2004 and 2050 for water limited (WL) and nitrogen–water limited (NWL) production situations, under current management (Table 1). Averaged across all crops and production situations, impacts were relatively more positive for the B1 scenario, and relatively most negative for the B2 scenario. Among the main crops, relative yield impacts were most positive for the winter sown cereals (wheat, barley and rapeseed) (expect for the NWL simulations in the B2 scenario), reflecting the combined benefit of CO_2 fertilization for C3 crops and warmer temperatures in Northern locations. Grain maize and potato were expected to experience yield declines due to climate change to 2050 across all scenarios and production situations, due to shortened growth periods, too high temperatures for tuber filling for potato, and more severe drought periods. For all crop and scenario combinations, NWL simulations resulted in more negative (grain maize and potato) or less positive (wheat, barley and rapeseed) relative yield changes (Δ_{NWL}) than WL simulations (Δ_{WL}). The ability of the two (WL and NWL) calibrated models to simulate crop yields for the baseline period is shown in Table 2 and Fig. 3. On average, the WL yields were 37% higher than the NWL yields, with a range of 10% (sugar beet) to 57% (winter barley). In all cases, WL yields were higher than observed yields, while NWL yields were higher than observed yields in all cases except for winter barley.

The European aggregates hide a great deal of variability across simulation units as well as national boundaries. The actual impact of climate change depends largely on the combination of water availability, temperatures at various crop development stages, crop thermal requirements and management, and is best considered at a more dissaggregated scale. The spatial distribution across Europe of climate and CO_2 impacts are shown for Δ_{WL} simulations in Fig. S2 and for Δ_{NWL} in Fig. S3, in both cases for the A1B scenario. The difference between the WL and NWL simulations were small for

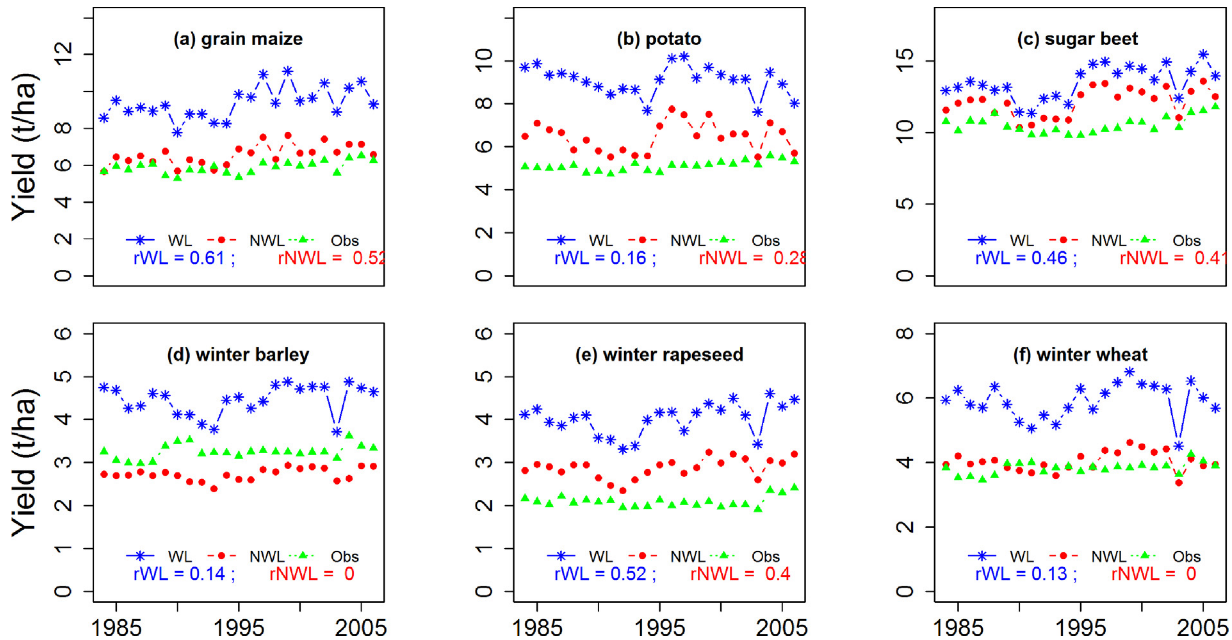

Fig. 3. European average annual time series of observed and water limited and nitrogen water limited simulated yields for grain maize (a), potato (b), sugar beet (c), winter barley (d), winter rapeseed (e) and winter wheat (f). The p values at the bottom of the plots are the Spearman's correlation coefficients between simulated (from left to right, simulated-water limited and simulated nitrogen water limited) and observed yields.

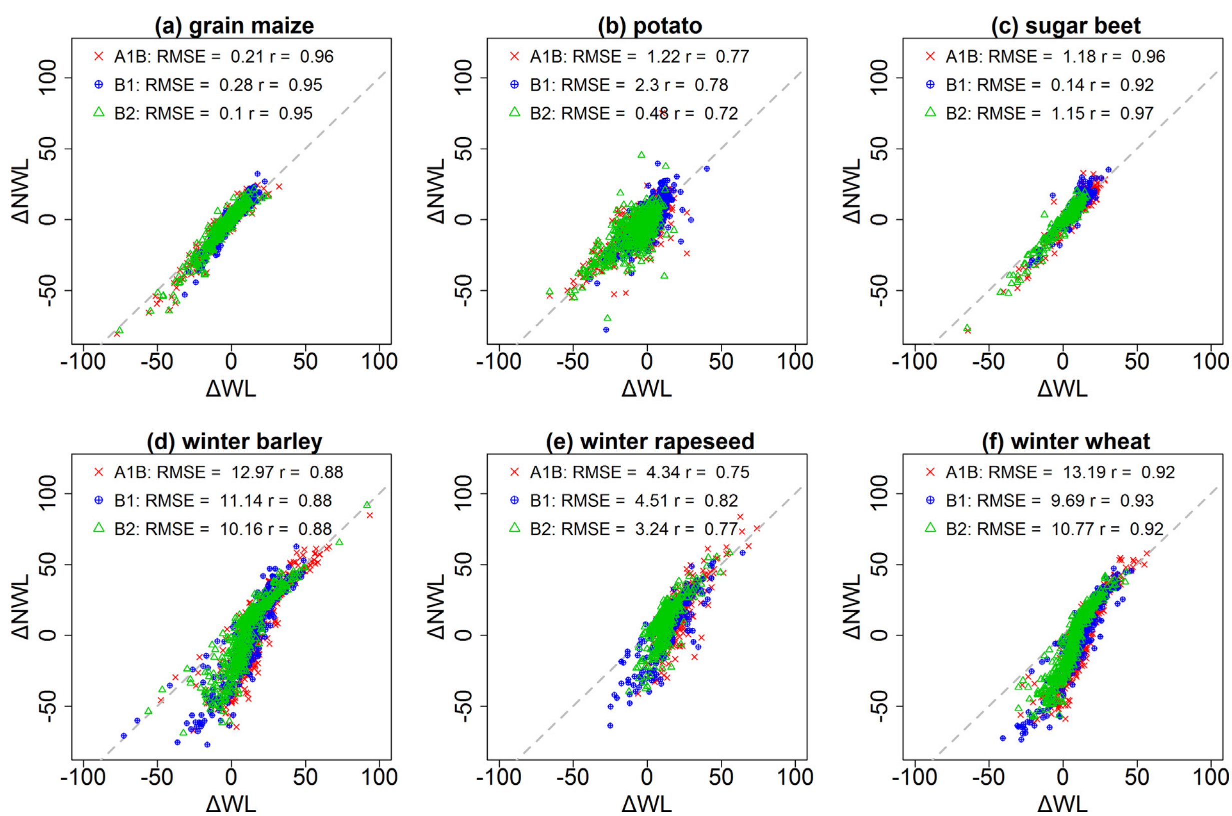

Fig. 4. Correlation between relative changes in crop yield between 2004 and 2050 for WL (Δ_{WL}) and NWL (Δ_{NWL}) simulations and production situations for the six crops for the three climate change scenarios across Europe.

grain maize and sugar beet, intermediate for potato and relatievly large for winter barley, winter rapeseed and winter wheat for all scenarios (Fig. 4). For the latter three crops, Δ_{NWL} was generally more negative than Δ_{WL} particularly for Southern Europe. $\Delta\Delta$ had the largest magnitude in Spain for winter barley, winter rapeseed and winter wheat (Fig. 5). These three crops are all largely non-irrigated (Zhao et al., in press) in the region, and Δ_{WL} in these regions for winter cereals tended to be more negative than other regions.

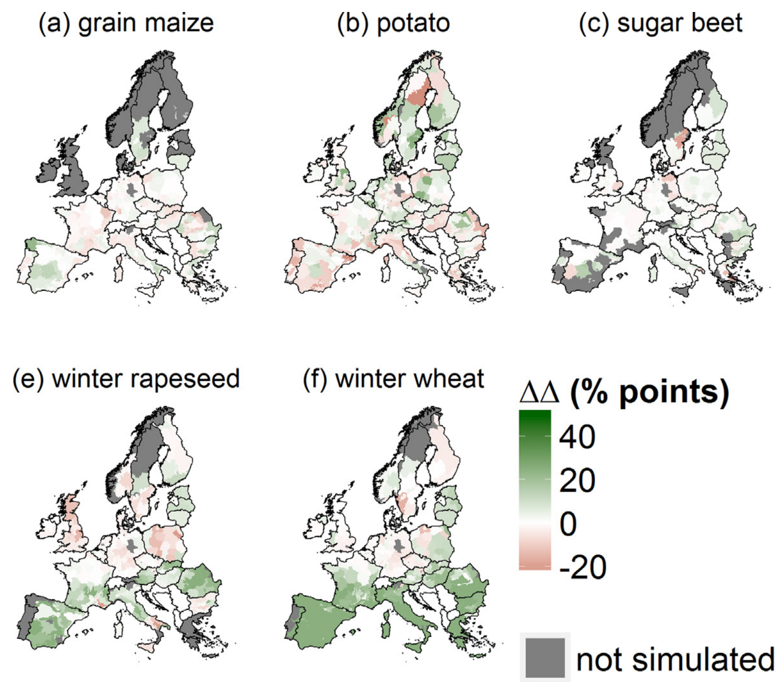

Fig. 5. Difference between the relative yield changes between 2004 and 2050 for WL (Δ_{WL}) and NWL (Δ_{NWL}) simulations and production systems, $\Delta\Delta = \Delta_{WL} - \Delta_{NWL}$, for the six crops and the A1B scenario across Europe.

3.2. Potential change in nitrogen demand with climate change

The combined impact of climate change and elevated CO_2 concentrations on crop nitrogen uptake, expressed as relative changes between 2004 and 2050, $\Delta_{NU,WL}$ and $\Delta_{NU,NWL}$, for WL and NWL production situations respectively, are summarized in Table 3. The spatial distribution of $\Delta_{NU,WL}$ and $\Delta_{NU,NWL}$ across Europe are shown in Fig. S4 and S5, for WL and NWL simulations, respectively for the A1B scenario. Averaged across all crops and both WL and NWL production situations, nitrogen use was simulated to increase by 3% in the A1B and B1 scenarios, and remained constant in the B2 scenario. However, if only winter sown crops are considered, nitrogen use was expected to increase by 10% on average across all scenarios, with the largest increase of 14% expected with the A1B scenario. When averaged across Europe, $\Delta_{NU,NWL}$ was more negative than $\Delta_{NU,WL}$ for all crops. For the winter sown crops, $\Delta_{NU,WL}$ was large and positive in the South and decreased moving northward (Fig. S4). The opposite spatial pattern existed for $\Delta_{NU,NWL}$ (Fig. S5). For the spring sown crops, patterns in $\Delta_{NU,WL}$ and $\Delta_{NU,NWL}$ largely mirrored one another, with $\Delta_{NU,NWL}$ being less positive.

For grain maize and sugarbeet, correlations between the relative changes in future nitrogen uptake were high between the two production situations WL and NWL, but were poor to moderate for the winter crops (Fig. 6). This pattern was consistent across the three climate scenarios. Likewise, the spatial distribution of $\Delta\Delta_{NU}$ for the spring sown crops (except for potato) show that there were relatively small differences between $\Delta_{NU,WL}$ and $\Delta_{NU,NWL}$ for spring crops such as sugarbeet with larger differences for the winter crops (Fig. 7). For the winter sown crops, future nitrogen uptake generally increased for the WL simulations, particularly in Southern Europe (Fig. S4), but this was often not the case for the NWL simulations (Fig. S5). For example for winter wheat, the WL simulations showed a strong increase in nitrogen uptake in Spain, but reductions in large parts of Northeast Europe. However, the NWL simulations showed smaller differences in the nitrogen uptake across Europe (Fig. S5), resulting in differences in the spatial patterns of $\Delta_{NU,WL}$ and $\Delta_{NU,NWL}$ across Europe (Fig. 7).

4. Discussion

The aim of this study was to assess the need to consider nitrogen limitation for the analysis of climate change impacts on European cropping systems. Previously, Rosenzweig et al. (2014) used seven different global gridded crop models to determine differences in impact magnitudes between models considering nitrogen limitation versus those which did not. Rosenzweig et al. (2014) concluded that models without nitrogen limitation generally have a more optimistic outlook for cropping under climate change. While this suggests the need to consider nitrogen limitation, the models assessed varied widely in their modelling approaches and accounting of nitrogen availability. Therefore, the current study used the same model, with and without nitrogen limitation, as the basis to examine how sensitive the results from climate change impact studies on future yields and nitrogen use are to nitrogen limitation.

4.1. Model performance and study uncertainties

SIMPLACE<LINTUL5, DRUNIR, HEAT> was able to simulate NWL yields that reproduced the spatial variability of actual yields across Europe and captured some of the temporal variation of average European yields, particularly for grain maize, sugar beet and winter rapeseed (Fig. 3). Inclusion of nitrogen limitation in the simulations led to yield levels that were more directly comparable to actual yields, despite the later additionally being limited by other nutrients, pests, weeds and disease (Table 2). As detailed in Zhao et al. (in press), each simulation was calibrated for the phenology thermal time parameters to ensure that simulated anthesis and maturity matched observations in the baseline period following an approach used in Therond et al. (2011). However, despite the model's performance in the baseline, some important factors related to crop and soil nutrient dynamics were not considered in this study due to a number of uncertainties for regional scale crop modeling under climate change (Olesen et al., 2007). Warmer soil temperatures associated with climate change lead to faster soil nitrogen mineralization rates (Melillo et al., 2002). However, at a continental

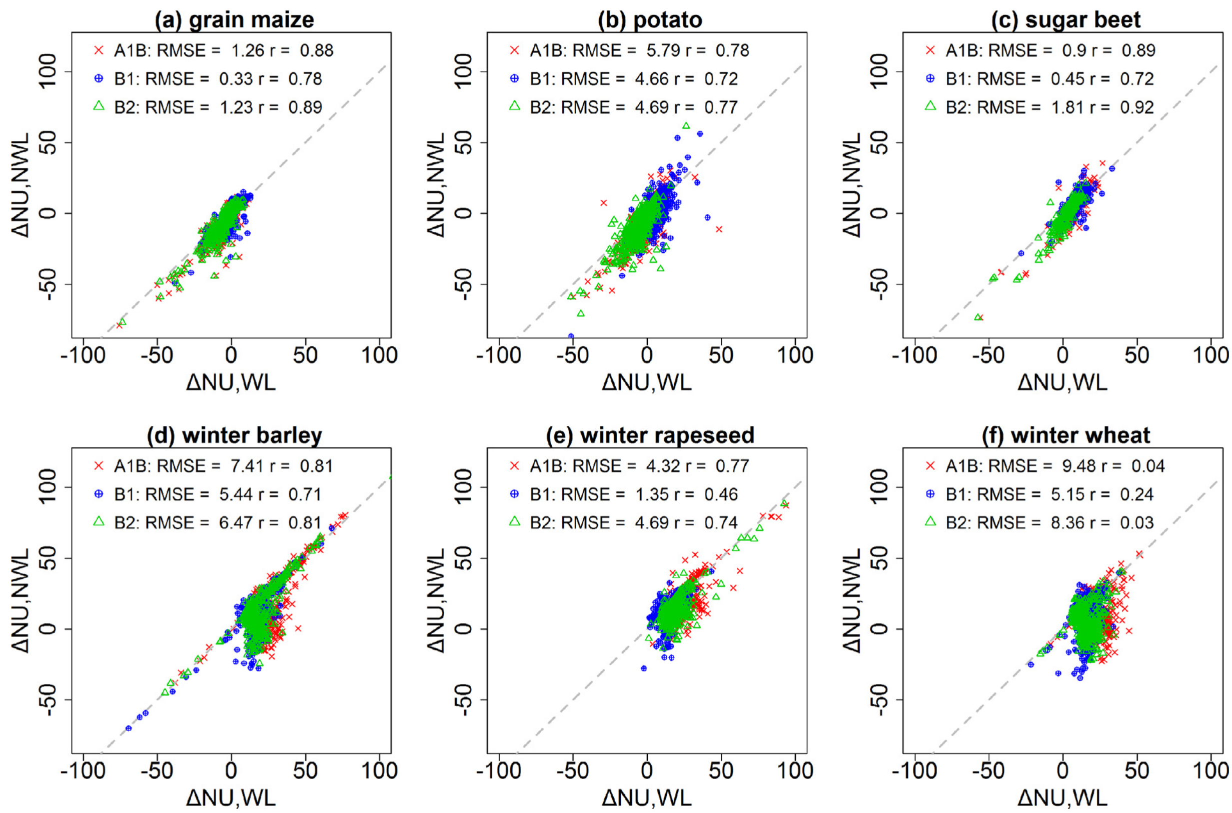

Fig. 6. Correlation between relative changes in crop nitrogen uptake between 2004 and 2050 for WL ($\Delta_{NU,WL}$) and NWL ($\Delta_{NU,WL}$) simulations and production systems for the six crops for the three climate scenario across Europe.

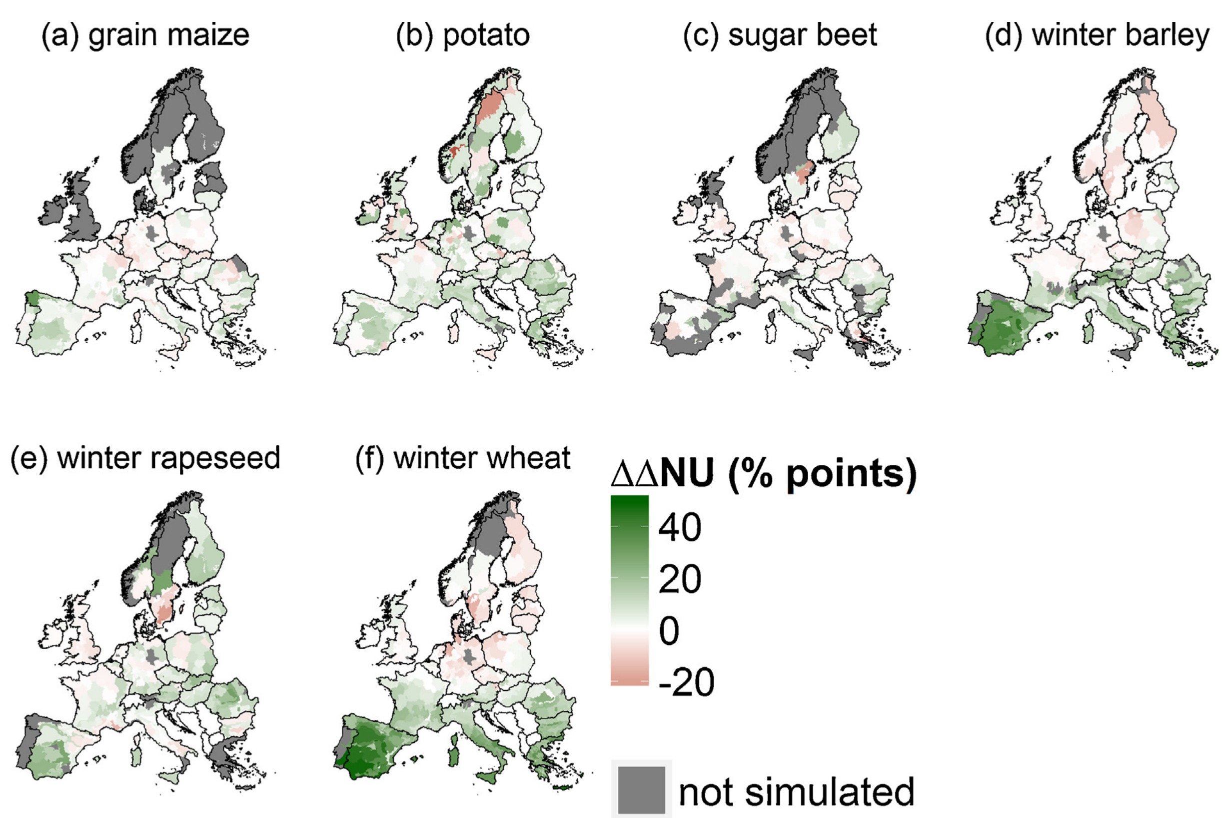

Fig. 7. Difference between the relative N uptake changes between 2004 and 2050 for WL (Δ_{WL}) and NWL (Δ_{NWL}) simulations and production systems, $\Delta NU,_{prod\_syst} = \Delta_{NU,WL} - \Delta_{NU,NWL}$ for the six crops and the A1B scenario across Europe.

Table 3
European aggregate relative changes in crop nitrogen uptake, $\Delta_{NuptakeWL}$ and $\Delta_{NuptakeNWL}$, estimated using water limited (WL) and nitrogen–water limited (NWL) simulations respectively, between 2004 and 2050 for three climate scenarios.

Crop	Relative change in crop nitrogen uptake (%)					
	A1B scenario		B1 scenario		B2 scenario	
	$\Delta_{NuptakeWL}$	$\Delta_{NuptakeNWL}$	$\Delta_{NuptakeWL}$	$\Delta_{NuptakeNWL}$	$\Delta_{NuptakeWL}$	$\Delta_{NuptakeNWL}$
Grain maize	−5	−4	−1	2	−6	−5
Potato	−2	−6	5	−1	−6	−9
Sugar beet	9	11	10	11	3	3
Winter barley	26	6	17	3	19	1
Winter rapeseed	25	15	14	10	19	8
Winter wheat	21	0	15	4	14	−4

scale, mineralization rates are also sensitive to soil C and N contents and precipitation (Colman and Schimel, 2013). We did not change mineralization rates but kept them constant in our regional scale crop model configuration. Likewise, we did not include the dynamics of nitrogen losses due to leaching, which other studies have shown will be impacted by climate change (Olesen et al., 2007; Thaler et al., 2012; Eitzinger et al., 2013), though not due to warming (Melillo et al., 2002). Olesen et al. (2007) found that with optimal nitrogen fertilization rates for winter wheat production, nitrogen leaching would decrease with climate change for large parts of Europe, though parts of North-west Europe would potentially experience greater leaching. However, the results were very sensitive to choice of regional climate model, scenario, location and soils. In a study examining nitrogen leaching under climate change in Austria for winter wheat and spring barley, Eitzinger et al. (2013) found N-leaching would almost double in 2035 for winter wheat and increase by more than 40% for spring barley if irrigation increased to meet the increased water demand under climate change. There was a high degree of variability across soil types, with the highest absolute leaching rates expected for the lightest soils. Further, Thaler et al. (2012) found that crop management strategies such as minimum tillage and irrigation decrease or increase the degree of N-leaching under climate change for winter wheat in Austria. In a companion study to this current paper, Wolf et al. (in press) demonstrated the variability of changes in N-leaching under climate change across a range of European locations with expected increases in leaching in Denmark and Poland and reductions or little change in the Netherlands, France and Spain. The differences were largely explained by the different rates of N-inputs. Despite the uncertainty that arises from omitting N-leaching, its inclusion was felt to be outside of the scope of this study for a number of reasons noted in the studies above. SIMPLACE<LINTUL5, DRUNIR, HEAT> does account for the fact that not all nitrogen fertilizer applied will be recovered using a constant reduction factor which is not responsive to precipitation or any other soil dynamics (Shibu et al., 2010). Realistic simulation of N-leaching would require the use of another crop-soil model needing more detailed soil information for model parameterization (Gaiser et al., 2013) and perhaps simulation at a finer spatial resolution to capture more variability in management and soil types. We may also have possibly overestimated leaf nitrogen contents as theoretically leaf nitrogen content is expected to decrease under elevated CO_2 concentrations (Drake et al., 1997). Photosynthetic acclimation to elevated CO_2 after prolonged exposure is caused by a selective reduction in Rubisco levels (Ainsworth and Long, 2005; Leakey et al., 2009), which constitutes a large fraction of leaf nitrogen amounts (Sage et al., 1987). However, in a meta-analysis of FACE experiments, Leakey et al. (2009) found less than a 4% reduction in leaf nitrogen levels for crops at elevated CO_2 and rather attribute higher photosynthetic nitrogen use efficiency (Drake et al., 1997) to gains in CO_2. Due to the minor effect and a lack of crop specific information, we did not include this in our model simulations.

An additional source of uncertainty in this study is related to the assumptions and methods used to determine future fertilizer nitrogen amounts for this study. The explicit and arbitrary assumption was made that management intensity would not change, or alternatively stated, that the yield gap due to nitrogen would stay the same in the future. Therefore, nitrogen application in 2050 would be adjusted from 2004 levels by the same amount that WL yields changed, Δ_{WL}. An alternative to this assumption is to assume that nitrogen fertilization rates do not change between the baseline and future time periods, as appears to have been done in the Rosenzweig et al. (2014) study. However, we elected to vary N_{2050} with the expected change in WL yields in the same manner as done in other studies to keep irrigation intensity the same (Eitzinger et al., 2013; Zhao et al., in press). We reasoned that farmers who currently fertilizer to reach WL yield levels would be likely to apply enough fertilizer in 2050 to reach WL yield levels in 2050, and not maintain fertilizer rates from 2004. In reality, fertilizer levels will probably be determined by considering input and product prices (Olesen et al., 2007; Wolf et al., in press). Further, as no consistent data was available on the timing of fertilizer application, all fertilizer was applied at the sowing, though producers, particularly for winter sown crops, are unlikely to follow such a practice. Had part of the nitrogen been applied in the spring, nitrogen limitation may have been reduced at anthesis. However, these issues highlight both the challenge of approximating future nitrogen fertilizer rates and the uncertainty in present nitrogen input data (Ewert et al., in press). For example, it is not possible to know if all fertilizer bought in one year applied is in the same year, nor was it possible to distinguish between different manure and residues types or the timing of the various applications.

Another potential source of uncertainty is in the specification of the initial soil nitrogen values. While we did not perform a sensitivity analysis of the impact of this assumption on $\Delta\Delta$, there was a relationship between the size of $\Delta\Delta$ and the historical nitrogen inputs, but only for the winter crops in Southern Europe (Fig. A6, results for the spring crops not shown). These regions tend to have low rates of N inputs (Fig. A1) which were further decreased due to negative values of ΔWL. This suggests that $\Delta\Delta$ is sensitive to initial soil nitrogen values, though it is unclear how this could be improved without access to improved soil information across Europe.

4.2. Effect of N-limitation on simulated relative yield changes and nitrogen demand

Simulated yield changes, Δ_{WL} and Δ_{NWL}, were generally comparable for spring sown crops, but large yield change differences occurred for the winter sown crops. This was particularly pronounced in Spain and Italy, due to the strong yield decreases in NWL simulations (Fig. 5). To understand what drives the difference between Δ_{WL} and Δ_{NWL} recall that changes in Δ_{WL} are used to determine N_{2050}. If Δ_{WL} is negative, less nitrogen will be applied in the future. If negative values of Δ_{WL} mainly resulted from short-

Table 4
European aggregate relative changes in the length of the growing season between 2004 and 2050.

Crop	Relative change in days in growing season (%)	
	Mean	Standard deviation
Grain maize	-22	4.3
Potato	-12	2.8
Sugar beet	-15	4.9
Winter barley	-5	0.7
Winter rapeseed	-5	0.4
Winter wheat	-5	0.7

ened growing seasons due to accelerated crop development, an equivalent reduction in nitrogen use seems logical and Δ_{WL} and Δ_{NWL} should be approximately the same. This was the case for the simulations of the spring sown crops. Most of the simulation units showed that the growing season was shortened to a much greater extent for the spring sown crops than for the winter sown crops, with the former exhibiting greater variability (Table 4). For the spring sown crops, the growing season length is reduced by between 22% for grain maize and 12% for potato, whereas the winter crops average less than a 5% reduction in growing season length. This phenomenon is explained by the buffering effect of vernalization and photoperiod effects acting on the winter sown crops, but not on the temperate spring sown crops (van Bussel et al., 2015). This implies that while the variation in the impact of WL yields across Europe (Fig. A2) for the winter sown crops is explained almost entirely by differences in water stress, the impacts of climate change on the WL yields of spring sown crops is determined by a combination of changes in drought stress and the growing season length.

$\Delta\Delta$ was generally positive (i.e. Δ_{NWL} was more negative than Δ_{WL}) for the winter crops in the Southern countries where nitrogen input rates are low (Fig S1). Examination of the relative increase in crop nitrogen uptake, $\Delta_{NU,WL}$, in the water limited simulations (Fig. S4) showed an increase in nitrogen uptake for the winter cereals in Spain and Italy despite negative Δ_{WL}. However, with a negative change in Δ_{WL} less nitrogen fertilizer was applied in the scenario runs than in the baseline period. The winter crops experienced warmer autumns in which they could produce more biomass, particularly under higher ambient CO_2 concentrations, without shortening their growth cycle as explained above due to photoperiod effects and vernalization requirements. As a result, they took up more nitrogen in the winter season without accelerating development, with the result that less nitrogen remained for growth in the spring and summer during yield formation. In Southern Europe in the WL simulations, the benefit of more favourable autumn growing conditions and a greater capture of radiation did not translate into increased yield due to summer drought stress which occured around the sensitive time for yield formation. Drought stress increased in the scenarios for the largely non-irrigated winter crops in Southern Europe due to both higher temperatures and reduced precipitation and the larger LAI obtained in the autumn. For the spring sown crops, negative yield impacts on WL yields in the south were largely a function of shortened growing season as the spring crops are largely irrigated in Spain, Italy, Southern France and other eastern countries (see Zhao et al., 2015a). Therefore, while nitrogen application rates were decreased in southern Europe for these crops, the nitrogen demand was reduced by approximately the same amount due to the shorter growing season (Fig. A5), with the result that the impact of climate change on the yields of the spring sown crops was equivalent for WL and NWL simulations.

It is not clear if this phenomenom of limited reductions in the growing season length for the winter crops explains the more negative yield impacts of N limited simulations reported in Rosenzweig et al. (2014) as most of the models in their study include vernalization but not photoperiod effects on phenology development. Also, they found that grain maize experienced more negative climate change impacts with N limitation which our study did not. However, their study is global and for the models considering photoperiod effects, maize may have been parameterized as being a short day plant as it is in tropical regions, whereas it is day-neutral in temperate regions like much of Europe. Finally, it is not clear how future N rates were specified in their study.

The outcome of our study is that consideration of N limitation is important for understanding how winter sown crop yields and N use might change with climate change. However, our results suggested that consideration of N use was not critical to assess climate change impacts on spring sown crops. Finally, as recently demonstrated by Wolf et al. (in press), future yields are expected to be much more strongly impacted by technology developments, through either increasing yield potential or decreasing yield gaps, than by climate change and these results should be considered in that context.

4.3. Change in nitrogen demand under climate change

Regarding future N demand, our results suggested that at a European scale and averaged across all crops, N rates would not change much in the future if current management intensity, sowing date and varieties were maintained. However, large differences existed between crops and regions. Across all regions and scenarios, nitrogen use was mainly expected to decrease for spring sown crops (except for sugar beet). The explanation lies with the fact that climate change would lead to faster crop development rates and lower yields, with little benefit of elevated CO_2 concentrations for C4 crops. This finding for spring crops was largely independent of whether one considers WL or NWL production conditions (Fig. 6), and no striking patterns were apparent across Europe. Future nitrogen use rates for the winter sown crops were more spatially variable and differ widely with the production situation considered (Fig. 7). Under WL production systems, N use would increase in the South and decrease in parts of Northern Europe. However, under the more realistic NWL production system, though with high uncertainty in the N inputs, spatial patterns differed across Europe with respect to future nitrogen use.

While our study demonstrated that consideration of N-limitation is needed when simulating climate change impacts on winter crops, research is needed to explore the implications of different methods to estimate future nitrogen use. The impacts of terminal drought on the non-irrigated winter cereals in Spain and Italy which lead to only small increases, or negative changes, in water limited yields, also produced lower harvest indices. Therefore, consideration of relative change in biomass may have been a better estimator of future N rates. However, farmers are unlikely to increase fertilizer rates to only achieve more biomass, and specification of future fertilizer rates will need to consider likely adaptation (Donatelli et al., 2012) and be determined simultaneously with crop and economic models, within the framework of an integrated modeling study. However, the exercise is expected to be complicated as actual N rates are strongly dependent on the sowing date and timing of nitrogen applications. This study applied all nitrogen at sowing to ensure consistency across simulation units. As this is not representative of actual farmers' practices, future study should investigate the sensitivity climate change impacts to the timing of fertilizer applications. Farmers are expected to mitigate some yield losses (and hence reductions in N application) by adjusting sowing dates and crop varieties (Donatelli et al., 2015). Finally, to date no studies have systematically examined the implications of aggregating and scaling N input management data and the yields used to calibrate models considering NWL conditions as typically found in reality.

5. Conclusions

Nitrogen fertilization is a key determinant of crop yield and varies widely across Europe, yet it is largely neglected in climate impact assessment studies. Our study simulated water-limited and nitrogen–water limited yields across the EU-27 for six crops with the goal to assess how conclusions about climate change impacts on crops depend on the consideration of nitrogen limitation. Our results suggested that for the spring-sown crops considered (grain maize, potato and sugar beet), the additional model complexity (parameterization, input data and assumptions about future use) of including N responsiveness did not change the crop response to climate change. However, for winter-sown crops (winter barley, winter rapeseed and winter wheat), simulated impacts to 2050 were more negative when N limitation was considered. This effect was particularly evident where water stress was significant. Future N use rates were likely to decrease due to climate change for spring-sown crops, largely in parallel with their yields. Future N use rates for winter-sown crops showed more spatial variability and may increase or decrease across Europe. Estimations of future N use were highly uncertain due to both uncertainty in current N use and its availability to the crop, and assumptions regarding how N use intensity might change in the future. We conclude that European climate change impacts assessments considering winter crops should account for possible limitations of N. We also see a demand for research to better understand and model present and future nitrogen rates and their effects on crops in Europe.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.eja.2015.09.002.

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